Four uncertainty objects

Same groups; different targets and laws

(a) Rank confidence set

Law: repeated sampling P

Data vary; rank $r_j(P)$ is fixed

$R_{n,j}(Y)$
Data-dependent rank set

One j: $r_j(P) \in R_{n,j}(Y)$

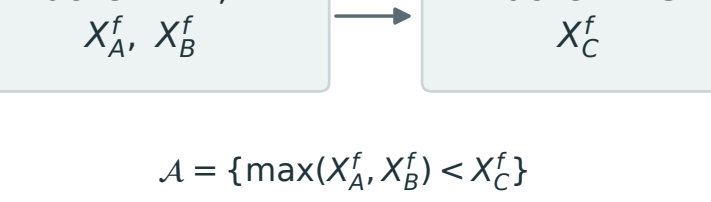
All j: $\forall j:\ r_j(P) \in R_{n,j}(Y)$

Marginal and simultaneous coverage are distinct event scopes.

(b) One posterior comparison

Law: posterior $\Pi_s(\cdot \mid y)$

Latent state varies; data y are fixed



$$E_{AB} = \{\theta_A > \theta_B\}$$

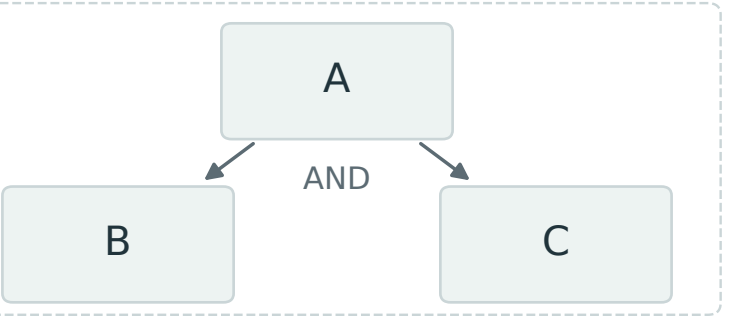
$$\Pi_s(E_{AB} \mid y)$$

C is not compared.

(c) A strict joint statement

Law: posterior $\Pi_s(\cdot \mid y)$

Latent state varies jointly; y is fixed



B versus C unspecified

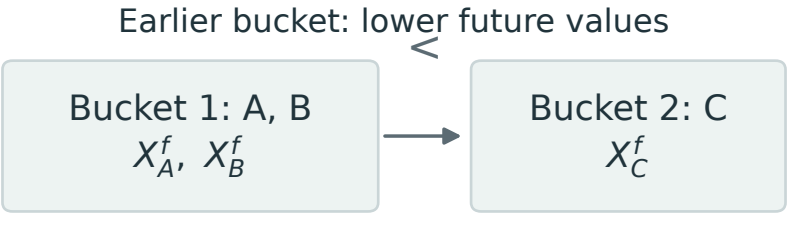
$$\Pi_s(E_{AB} \cap E_{AC} \mid y)$$

Strict conjunction example

(d) Future-observation buckets

Law: NPI predictive probabilities

One future observation varies per group



$$\mathcal{A} = \{\max(X_A^f, X_B^f) < X_C^f\}$$

$$P_{\text{low}}(\mathcal{A}), \quad P_{\text{up}}(\mathcal{A})$$

Within a bucket: order unspecified